

Hydrogeochemistry and Source Appraisal of Groundwater Quality in a Semi-Arid Region of Kadapa, India: Insights into Pollution, Health Risks

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ABSTRACT

Groundwater quality is a major concern in the semi-arid regions such as Badvel, India, where the geogenic, and climatic factors are compounded by intensive agricultural activities, which may exacerbate the contamination risk. To assess this, 50 groundwater samples were collected and analysed for their physicochemical characteristics in the pre- and post-monsoon periods of 2023. pH, electrical conductivity (EC), total dissolved solids (TDS), major ions, total hardness (TH), and total alkalinity (TA) in groundwater were assessed, and a subsequent ion balance error (IBE) was conducted. Then, the pollution index of groundwater (PIG) and human health risk assessment (HHRA) were performed using established procedures. Moreover, the ionic sources were evaluated using conventional graphical, statistical and PHREEQC 3.7.3 simulation. Among all physicochemical parameters, TDS, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , NO_3^- and F^- were excess in groundwater. PIG revealed that (90%) of groundwater fell in the low pollution category, followed by moderate pollution (10%). The hazard quotient (HQ) for NO_3^- and F^- in groundwater for the majority of samples was >1 , indicating potential health risks to infants, followed by children, and adults. Notably lower PIG and HQ values were observed in post-monsoon groundwater compared to the pre-monsoon groundwater owing to the dilution effect of freshwater recharge. Moreover, precipitation of carbonate, sulphate, fluoride, and dissolution of chloride minerals were responsible for groundwater chemistry. Furthermore, the dominance of Na-Cl water facies and rock-water interaction were major chemical processes of groundwater. The dominance of F^- in groundwater was attributed to the rock-water interaction, whereas the NO_3^- may be attributed to anthropogenic influences. Hence, prior water treatment and continuous monitoring are necessary to avoid potential health risks to residents of Badvel, India.

Keywords: Groundwater pollution, Pollution index of groundwater (PIG), Multivariate statistical analysis, PHREEQC simulation, Human health risk assessment (HHRA), Badvel.

INTRODUCTION

Water is a necessity for the survival and sustainable development of humans and ecosystems on Earth. The usability of water for diverse practices depends on its physical, chemical, and biological characteristics. Currently, the quality of this precious resource is being compromised by the interplay of geological, anthropogenic and climatic factors (Das et al., 2022, 2023; Tiwari and Rao, 2025). In general, the presence of all ions, metalloids in groundwater is primarily attributed to geochemical interactions between water and rocks. However, anthropogenic activities amplify the concentrations of ions and metalloids in water. Elements such as fluoride, chloride, sodium and metalloids like arsenic, manganese, selenium and iron are sourced from geological influences (Sreedevi et al., 2019). The elements, such as nitrates, phosphates, lead, zinc and mercury are mostly sourced from anthropogenic activities like the usage of chemical fertilizers in agriculture, inappropriate management of urban and industrial waste (Li and Wu, 2019). Climate change impacts, like increases in temperature and frequent floods, may lead to extreme droughts that make water scarce and spread waterborne diseases like typhoid, cholera, and malaria, respectively.

Globally, groundwater pollution by fluoride and nitrate is a major public health concern nowadays. More than 23 countries worldwide have higher fluoride levels, including South America, China, India, and Central Africa, which grapple with alarming fluoride levels jeopardizing millions' health and livelihood. In general, the primary source of fluoride is attributed to geogenic sources, notably fluorite and fluorapatite minerals found in granites, shales, and limestones (Shaji et al., 2018; Sunitha and Muralidhara Reddy, 2018; Gugulothu et al., 2022). Moreover, fluoride sources further include agrochemicals and industrial activities. In addition, nitrates are one of the major contaminants in water, and their sources primarily include anthropogenic activities such as leachates from untreated urban sewage and the intensive application of chemicals in agriculture. Its elevated levels may cause severe health problems like nerve damage, kidney damage and even carcinogenicity (Madhav et al., 2021). Hence, evaluating fluoride and nitrates and their detrimental health effects on humans is necessary to ensure safe drinking water standards and protect public health. Previously, several researchers, like Madhav et al. (2021), Reddy et al. (2022)

and Yarlanki et al. (2025), have worked on health risk assessment of nitrates and fluorides.

In the semi-arid agrarian region of Badvel, India, groundwater serves as the primary source of water for human consumption, urban needs, and agricultural purposes; nonetheless, pressing environmental concerns surround its dwindling availability, deteriorating quality, and precarious quantity. Therefore, regional and national authorities may find this study useful in informing their decisions and developing sustainable resource management plans. The residents of the Badvel area are facing water quality issues, especially from fluoride and nitrate toxicity. Previously, only a few studies had focused on fluoride toxicity in and around the vicinity of Badvel, India (Santiah et al., 2020; Sirigala et al., 2023). The toxicity of nitrates in groundwater remains uninvestigated. A significant knowledge gap exists in understanding the hydrogeology, source appraisal, and risk assessment for human health of groundwater in Badvel, India. To bridge this gap, a comprehensive groundwater assessment was undertaken, utilising an integrated, multi-disciplinary framework that combined multivariate statistical analysis, conventional graphical methods, geochemical simulation, pollution indexing and human health risk assessment. The United Nations Sustainable Development Goal 6 (SDG 6) aims to achieve universal access to safe water and sanitation for all. Moreover, it emphasizes continuous monitoring and evaluation to reduce pollution and effectively manage contaminants, thereby improving the quality of the water. In pursuit of SDG 6, the following objectives are proposed: a) To evaluate the groundwater quality of

Badvel, India, in compliance with the Bureau of Indian Standards. b) To evaluate the spatio-temporal variation of pollution levels, source appraisal using statistical, geochemical, conventional graphical, and health risk evaluation methods, with special focus on nitrate and fluoride contamination.

STUDY AREA

Badvel, India, is situated between $14^{\circ}49'1''$ and $14^{\circ}37'54''$ N latitudes and $78^{\circ}56'45''$ and $79^{\circ}6'36''$ E longitudes (Fig. 1). The total area of Badvel is approximately 146.66 km². Badvel experiences a semi-arid climate. The people in Badvel mainly depend on groundwater for various purposes. The lithology of the area comprises primarily shales and shales with limestones of the Cumbum Formation, which exhibit low porosity values but may develop secondary porosity through fractures and diagenesis, indicating moderate groundwater prospects (Reddy et al., 2000). The sections along the Sagileru, a tributary of the Penna River, are covered in recent alluvium deposits, which indicate good groundwater prospects (Reddy et al., 2000). The mean annual rainfall in Badvel, India, is 800 mm (CGWB, 2025). Agriculture is the primary land use category in Badvel, followed by fallow land, forests, urban settlements, and surface water. The major soils are red ferruginous soils, followed by traces of alluvium along the Sagileru River. Piezometric levels vary between 26 and 89 feet below ground level (bgl). The mean depth of borewells across the study area ranges from 76 to 213 m. The study area is facing perilous groundwater overdrafts to deal with the escalating needs of agricultural and household uses.

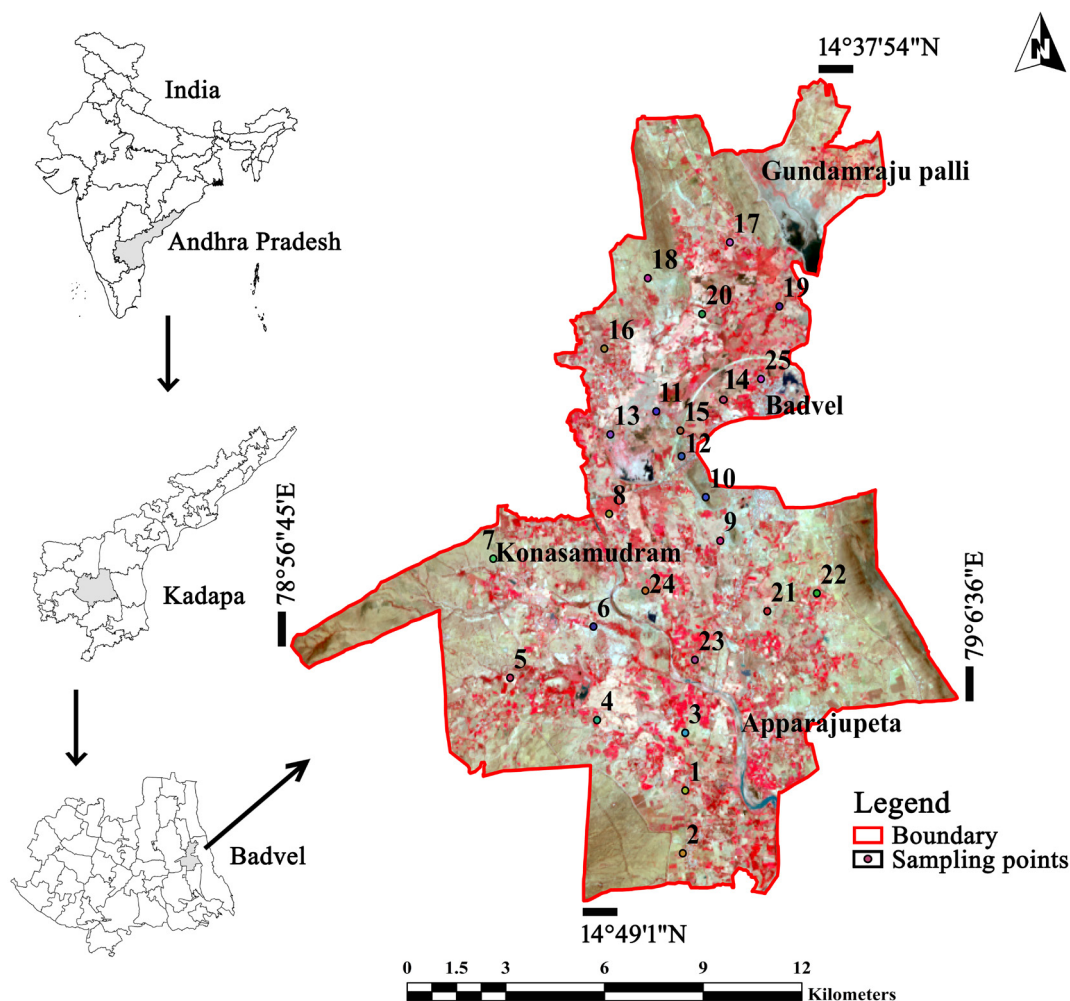


Fig. 1. LISS III satellite map of the study area with sampling points.

METHODOLOGY

Fifty groundwater samples were obtained in clean polyvinyl chloride (PVC) containers in the pre- and post-monsoon periods of 2023. pH, EC, and TDS were calibrated at the place of collection using a HI991300P meter. Nitrate (NO_3^-), fluoride (F^-), and sulphate (SO_4^{2-}) were measured using a UV-Visible spectrophotometer. Calcium (Ca^{2+}), magnesium (Mg^{2+}) and TH were analysed using an EDTA titrimetric procedure, and alkalinity parameters (HCO_3^- , CO_3^{2-} , and TA) were determined by acid-base titration using sulfuric acid (H_2SO_4) solution. The sodium (Na^+) and potassium (K^+) ions were quantified using a flame photometer (APHA, 2023). Later, the IBE was measured for analytical accuracy. Then, spatiotemporal patterns of PIG were portrayed in ArcGIS 10.4, and HHRA was calculated. Subsequently, the SI, cluster, Pearson's correlation, Piper, Gibbs, and scatter plots were portrayed. The procedure is discussed in detail in the following sections.

Ion-balance Error

In general, the sum of ions should be equivalent in solutions. However, the balance between ions is disrupted by analytical errors and unfiltered particulate matter. A positive IBE signifies dominance of cations, while a negative IBE signifies the dominance of anions. Hence, it is essential to know the accuracy. The IBE can be computed using the following formula.

$$\text{IBE} = \frac{\sum \text{Cations} + \sum \text{Anions}}{\sum \text{Cations} - \sum \text{Anions}} \times 100 \text{ meq} / \text{Lz} \quad (1)$$

An IBE of $\pm 10\%$ can be acceptable for water quality assessment (Yarlanki et al., 2025).

Water Quality Assessment and Pollution Index of Groundwater (PIG)

Fifty groundwater samples were gathered and analysed for physicochemical metrics using standard procedures. Then the analysed parameters were compared with BIS standards and assessed for pollution levels using PIG. The PIG developed by Subba Rao (2012) is an arithmetic index to calculate the degree of pollution in water. PIG assesses the physicochemical parameters of water in accordance with the standards established by the Bureau of Indian Standards (BIS, 2003). To determine the degree of pollution, the PIG evaluates the overall impact of pH, TDS, TH, TA, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , HCO_3^- , SO_4^{2-} , NO_3^- and F^- . The PIG of each water sample is evaluated using the standard formulations listed below in the phases indicated.

$$\text{Wp} = \frac{\text{Rw}}{\sum \text{Rw}} \quad (2)$$

In this context, $\sum \text{Rw}$ is the cumulative relative weight of all parameters, and Wp denotes the weight parameter. The Rw is a value assigned based on the influence of each parameter on human health, which varies between 1 and 5. Among the values, pH, TDS, TH, SO_4^{2-} , NO_3^- and F^- will be given 5, whereas Na^+ and Cl^- will be given 4, TA will be given 3, Ca^{2+} , Mg^{2+} , and HCO_3^- will be given 2, and K^+ will be given 1. In this context, 5 signifies a greater impact on human health, while 1 indicates a lesser effect.

$$\text{Sc} = \frac{\text{C}}{\text{Ds}} \quad (3)$$

In this context, Sc signifies the concentration status of a given parameter; C stands for the concentration of the analysed

parameter, and Ds denotes each parameters standard limit set by BIS (BIS, 2003).

$$\text{Aw} = \text{Wp} \times \text{Sc} \quad (4)$$

Aw denotes the aggregate water quality; Wp denotes the cumulative relative weight parameter, and Sc denotes the concentration status.

The PIG calculates the total of all Aw values for all parameters in each sample.

$$\text{PIG} = \sum \text{Aw} \quad (5)$$

The PIG is categorised into five levels, ranging from insignificant (<1) to very high pollution (2.5 to 3). These categories are: insignificant (<1), low (1 to 1.5), moderate (1.5 to 2), high (2 to 2.5), and very high pollution (2.5 to 3). The spatial and temporal dynamics of PIG were analysed using the inverse distance weighted (IDW) tool in ArcGIS 10.4.

Human Health Risk Assessment of Nitrates and Fluorides

In general, fluoride is one of the essential elements that aid in bone and teeth development. However, elevated levels of fluoride can lead to dental and skeletal fluorosis (Sunitha and Muralidhara Reddy, 2018). Fluoride in water is sourced from geological processes, including the weathering of fluorite and fluorapatite minerals found in granites, syenites, and shales (Shaji et al., 2023). Hence, its evaluation and potential detrimental health implications are crucial to ensure public safety. Moreover, nitrate is one of the major contaminants in water, which may be released from anthropogenic processes, including intensive agricultural practices and the inappropriate disposal practices of municipal and mining waste, along with natural processes, such as the degradation of nitrate minerals like nitratine and tobelite, and rainfall-induced nitrification. Excess NO_3^- in water can cause methemoglobinemia, thyroid disorders, renal problems, and an increased cancer risk (Madhav et al., 2021). Hence, its evaluation and potential health risks to humans are crucial to take proper remedial measures.

The nitrates and fluoride hazard quotient (HQ) for infants, children, and adults are calculated using the following formula.

$$\text{CDI} = (\text{CW} \times \text{IR} \times \text{EI} \times \text{ED}) / (\text{MBW} \times \text{MT}) \quad (6)$$

In this context, CDI signifies chronic daily intake (mg/kg-day); IR signifies ingestion ratio (adults: 2.5 L/day, children: 1.5 L/day and infants: 0.8 L/day); Cw for concentration of nitrate and fluoride in water (mg/L); EI signifies exposure interval (adults: 64 years, children: 6 years and 1.5 years for infants); ED signifies exposure duration (365 days); MBW signifies mean body weight (adults: 55Kg; children: 20g and infants: 10kg); MT signifies mean exposure time (adults: 23360 days; children: 2190 days and infants: 547 days); All these formulae and standards were based on standard procedures from USEPA, 2015.

$$\text{HQ} = \text{CDI} / \text{RD} \quad (7)$$

In this context, RD signifies the standard reference dose for human exposure to nitrates and fluorides (1.6 and 0.06 mg/kg/day, respectively). According to USEPA (2015), the permissible limit of HQ is 1. Exceeding this threshold may pose significant health risks to humans.

Source Appraisal

This study investigated statistical analysis, the saturation indices, conventional graphical representations such as piper, Gibbs diagrams, and scatterplots, along with advanced data reduction techniques, including cluster analysis and Pearson's correlation, to determine the dominant controlling mechanisms of groundwater in Badvel, India.

Statistical Analysis

In general, data reduction techniques such as cluster analysis and correlation matrix simplify complex datasets, revealing underlying factors and relationships. The cluster analysis and Pearson's correlation matrix were performed using MINITAB 22 software to identify the major governing processes in the groundwater of Badvel, India.

Saturation Indices (SI)

SI aids in predicting the dissolution or precipitation of minerals in aquatic habitats. In this study, the PHREEQC geochemical simulation, initially developed by the U.S. Geological Survey, allows for the instantaneous execution of multiple samples at a time.

The SI will be determined by using the following expression.

$$SI = \frac{\log IAP}{K_{sp}} \quad (8)$$

Here, SI >1 indicates precipitation, while the SI <1 signifies dissolving, whereas SI=0 denotes the balance of mineral species (Benavente et al., 2025).

Piper diagram

According to Piper (1944), the Piper illustration provides valuable insights into the dominance of water facies and rock-water interaction. Piper diagram, comprised of a central diamond plot, displaying four key ionic combinations, such as $\text{SO}_4^{2-} + \text{Cl}^-$ (upper left side), $\text{Ca}^{2+} + \text{Mg}^{2+}$ (upper right), $\text{Na}^+ + \text{K}^+$ (lower left) and $\text{CO}_3^{2-} + \text{HCO}_3^-$ (lower right side). Moreover, the Piper diagram illustrates a combination of two ternary diagrams: one represents $\text{Ca}^{2+} + \text{Mg}^{2+}$ (lower left) and the other represents $(\text{SO}_4^{2-} + \text{Cl}^-)$ in (lower right). The Piper illustration comprised six fundamental water facies: Ca-Mg-

HCO_3^- , mixed Ca-Na- HCO_3^- , Na-Cl, Ca-Cl, Na- HCO_3^- , and mixed Ca-Mg-Cl.

Gibbs and Scatter plots

According to Gibbs (1970), the chemistry of water is influenced by several factors, primarily evaporation, rock dominance, and precipitation. The Gibbs diagram effectively illustrates these processes, highlighting the complex interactions between water composition, geological settings, and climatic conditions. The Gibbs ratios I & II were calibrated using the formulae given below:

$$\text{Gibbs ratio I} = \frac{(\text{Cl}^-)}{(\text{Cl}^- + \text{HCO}_3^-)} \quad (9)$$

$$\text{Gibbs ratio II} = \frac{(\text{Na}^+ + \text{K}^+)}{(\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+})} (\text{meq/L}) \quad (10)$$

Additionally, to Gibbs, scatter plots of $(\text{Na}^+ \text{ vs } \text{Cl}^-)$, $(\text{Ca}^{2+} \text{ vs } \text{SO}_4^{2-})$, $(\text{Total cations vs } \text{Ca}^{2+} + \text{Mg}^{2+})$, and $(\text{EC vs } \text{NO}_3^-)$ were depicted to find their ionic sources.

RESULT AND DISCUSSION

Ion-balance Error

Findings indicate that the precision of the analysis falls within the limit of $\pm 10\%$ (Adimalla, 2019; Yarlanki et al., 2025).

Water Quality Assessment and PIG

Physicochemical Parameters

The pH in groundwater varied between 7 and 9, averaging at 8 in the pre-monsoon season, whereas the post-monsoon groundwater samples varied between 7.5 and 8.5, averaging at 8.1, indicating a slightly alkaline nature. All the water samples adhere to the BIS permissible levels, making them potable (Table 1). The pre-monsoon EC values varied between 720 and 3400, averaging at 1719 ($\mu\text{mhos/cm}$). Instead, EC values in post-monsoon varied between 700 and 3500, averaging at 1670 ($\mu\text{mhos/cm}$). The TDS values in pre-monsoon water varied from 340 to 2300, with a mean of 778 (mg/L). In contrast, the TDS values varied between 250 and 2200, with a mean of 747 (mg/L) in the post-monsoon water. Elevated EC and TDS

Table 1. Descriptive statistics of various components of groundwater for drinking according to the guidelines of BIS (2003).

S. No.	Constituents	Pre-monsoon			Post-monsoon				Pre-monsoon	Post-monsoon
		Min	Max	Mean	Min	Max	Mean	LTL ¹	SSLTL ²	SSLTL ²
1	pH	7	9	8	7.5	8.5	8.1	6.5-8.5	0	0
2	Electrical conductivity ($\mu\text{mhos/cm}$)	720	3400	1719	700	3500	1670	NG ³	/	/
3	Total dissolved solids (TDS) (mg/L)	340	2300	778	250	2200	747	500	18	18
4	Total hardness (TH) (mg/L)	30	120	59	25	92	53	300	/	/
5	Total alkalinity (TA) (mg/L)	74	260	185	75	250	177	200	12	7
6	Calcium (Ca^{2+}) (mg/L)	50	240	98	40	220	93	75	19	18
7	Magnesium (Mg^{2+}) (mg/L)	35	150	51	32	120	51	30	25	25
8	Sodium (Na^+) (mg/L)	230	540	392	210	550	383	200	25	25
9	Potassium (K^+) (mg/L)	12	90	27	13	80	24	12	24	25
10	Carbonate (CO_3^{2-}) (mg/L)	0	32	8	0	32	8	NG ³	/	/
11	Bicarbonate (HCO_3^-) (mg/L)	70	240	177	100	250	183	200	7	8
12	Chloride (Cl^-) (mg/L)	100	290	186	90	270	177	250	3	1
13	Sulphate (SO_4^{2-}) (mg/L)	120	350	194	100	340	188	200	9	8
14	Nitrate (NO_3^-) (mg/L)	45	80	60	40	72	53	45	23	17
15	Fluoride (F^-) (mg/L)	1.3	4.4	2.5	1.2	4.2	2.3	1	25	25
16	Gibbs ratio – I (meq/L)	0.44	0.85	0.64	0.41	0.81	0.09	NG ³	/	/
17	Gibbs ratio – II (meq/L)	0.54	0.9	0.78	0.55	0.91	0.05	NG ³	/	/

¹Lower threshold limit: ²Samples surpassing lower threshold limit: ³No Guidelines.

concentrations in the groundwater of Badvel, India indicate that the water is significantly mineralized and has an extended residence time in aquifers. Notably, 72% of the groundwater exceeded the BIS limits in both monsoon seasons, making it not potable (Table 1). In Badvel, the TH values in groundwater vary between 30 and 120, averaging at 59 (mg/L), and from 25 to 92, averaging at 53 (mg/L) in the pre- and post-monsoon seasons, respectively. All the samples comply with the permissible limits set by BIS. The TA values varied from 74 to 260, averaging at 185 (mg/L), and from 75 to 250, averaging at 177 (mg/L) in the pre- and post-monsoon seasons, respectively. According to Kharaka et al. (2018), the TA in groundwater is primarily sourced from the dissolution of atmospheric CO₂ and calcite minerals present in rocks.

In Badvel, India, the pre-monsoon groundwater Ca²⁺ levels ranged from 50 to 240, with a mean of 98 (mg/L), whereas the Ca²⁺ values in groundwater varied between 40 and 220, with a mean of 93 (mg/L) in the post-monsoon season. Approximately 74% of the groundwater samples exceeded the BIS criteria, and about 26% were within the BIS criteria. The calcite mineral is the primary source of elevated levels through dissolution. The Mg²⁺ concentrations were between 35 and 150, averaging at 51 (mg/L); and from 32 and 120, averaging at 51 (mg/L) in the pre- and post-monsoon seasons, respectively. All the samples had values above the BIS standards (Table 1). Elevated levels of Mg²⁺ might be due to the weathering of shales containing minerals such as chlorite and magnesite, as well as an ion exchange process. In pre-monsoon groundwater, the Na⁺ ionic concentrations ranged from 230 to 540, averaging at 392 (mg/L), while they ranged from 210 to 550, averaging at 383 (mg/L) in post-monsoon groundwater. The K⁺ in the pre-monsoon groundwater varied between 12 and 90, averaging at 27 (mg/L), whereas in the post-monsoon season groundwater, they varied between 13 and 80 averaging 24 (mg/L). All the water samples had elevated levels of Na⁺ and K⁺, probably released owing to rock-water interaction and anthropogenic activities (Saravanan et al., 2016). The CO₃²⁻ concentrations in the pre-monsoon season, varied between 0 and 32, with a mean of 8 (mg/L), while in the post-monsoon season CO₃²⁻ concentrations also varied between 0 and 32, with a mean of 8 (mg/L). In the pre-monsoon season, groundwater HCO₃⁻ values varied from 70 to 240, averaging at 177 (mg/L); in contrast, the post-monsoon groundwater ranged between 100 and 250, averaging at 183 (mg/L) (Table 1). Nearly 30% of the groundwater samples had excess levels of HCO₃⁻ which comply with BIS standards. The source of HCO₃⁻ in water may be released from the weathering of CaCO₃ and MgCO₃ minerals in aquifers and freshwater recharge.

In Badvel, the Cl⁻ concentration varied between 100 and 290, averaging at 186 (mg/L), and between 90 and 270, averaging at 177 (mg/L), in the pre- and post-monsoons, respectively. In the pre-monsoon season, the SO₄²⁻ values in groundwater varied between 120 and 350, averaging at 194 (mg/L). While in the post-monsoon seasons, the SO₄²⁻ values varied between 100 and 340, averaging at 188 (mg/L). Further, 8% of Cl⁻ and a striking 34% of SO₄²⁻ exceeded the BIS limits. The elevated levels of SO₄²⁻ in groundwater may be attributed to the use of fertilisers, pesticides, and atmospheric precipitation. For nitrates, the pre-monsoon groundwater ranged from 45 to 80, with an average of 60 (mg/L), while the post-monsoon groundwater ranged from 40 to 72, with a mean of (53 mg/L). For pre-monsoon groundwater, F⁻ concentrations varied between 1.3 and 4.4 with a mean of 2.5 (mg/L), whereas for post-monsoon groundwater, F⁻ concentrations varied between 1.2 and 4.2 with an average of 2.3 (mg/L). 80% groundwater samples for nitrates and 100% of groundwater samples for fluorides exceeded the BIS limits in both pre- and post-monsoon seasons. The fluorides are of geogenic origin. Nitrates are of anthropogenic origin (Rajmohan and Elango, 2006; Brindha et al., 2012).

Pollution Index of Groundwater (PIG)

The PIG is a simple and trustworthy arithmetic index that several scholars have adopted for determining pollution levels in water (Subba Rao, 2012; Adimalla, 2019; Yarlanki et al., 2025). The result showed that 84% of groundwater showed low pollution and 16% of the groundwater exhibited moderate pollution, in the pre-monsoon season (Table 2). In contrast, during the post-monsoon season, groundwater showed moderate pollution in 4% of samples and low pollution in 96% of samples. Notably, 90% of the groundwater samples exhibited the low pollution category in both the pre- and post-monsoon seasons. Few groundwater samples shifted into the lower pollution category in the post-monsoon period, which could be attributed to the dilution process of freshwater recharge. (Figs 2a-b, 3).

Table 2. Pollution index of groundwater categories (Subba Rao, 2012).

PIG Range	Pollution category	Pollution percentage	
		Pre-monsoon	Post-monsoon
P<1	Insignificant	0	0
1-1.5	Low	84	96
1.5-2.0	Moderate	16	4
2.0-2.5	High	0	0
>2.5	Very High	0	0
		100	100

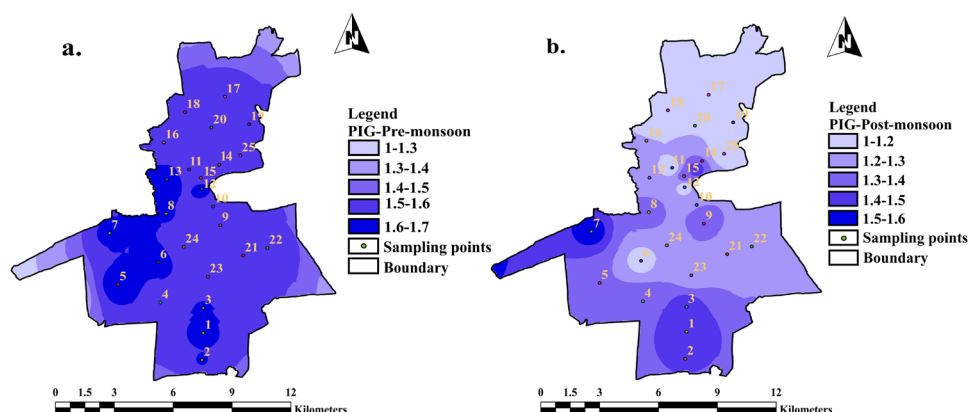


Fig. 2a -b. Spatiotemporal pattern of pollution index of groundwater (PIG).

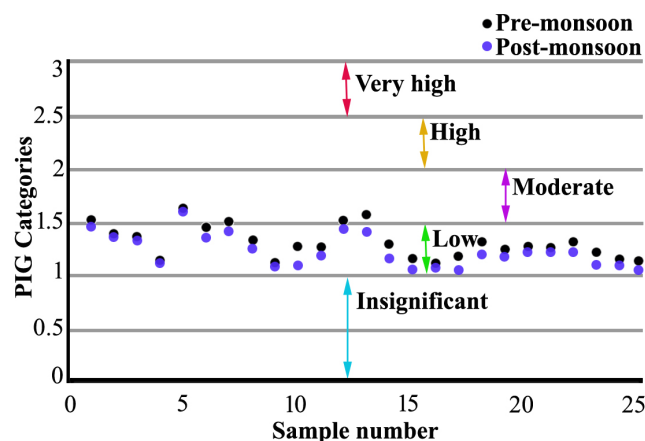


Fig. 3. Temporal trend of pollution index of groundwater (PIG).

Human Health Risk Assessment of Nitrates (NO_3^-) and Fluorides (F^-)

Nitrate and fluoride are major pollutants found in water, which pose potential health risks to humans; therefore, it is crucial to assess their impact on human health. The results reveal that, in Badvel, 100% of infants, 100% of children, and 52% of adults have surpassed the threshold value (>1) in the pre- and post-monsoon seasons (Tables 3 and 4 and Figs 4a-b and 5a-b). For fluoride, 100% of infants, 100% of children, and 32% of adults have shown raised threshold values (>1) in the pre- and post-monsoon seasons, which demonstrates that infants and children are at higher risk of nitrate and fluoride contamination than adults.

Source Appraisal

Saturation Indices (SI)

The SI of carbonate minerals (Aragonite, Calcite, and Dolomite), sulphate minerals (Gypsum, and Anhydrite), fluoride minerals (Fluorite) and chloride minerals (Sylvite, and Halite) in groundwater

Table 3. Summary statistics of nitrate hazard quotient.

S. No.	Pre-monsoon			Post-monsoon		
	Adults	Children	Infants	Adults	Children	Infants
1	0.8	2.1	2.3	0.8	2.0	2.1
2	0.9	2.4	2.6	1.0	2.6	2.8
3	0.9	2.3	2.5	0.9	2.4	2.6
4	0.8	2.1	2.3	0.7	1.9	2.0
5	1.3	3.3	3.5	1.3	3.4	3.6
6	0.9	2.3	2.5	0.8	2.1	2.3
7	1.0	2.6	2.8	0.8	2.1	2.3
8	1.0	2.7	2.9	0.8	2.0	2.1
9	0.9	2.3	2.4	0.8	2.1	2.3
10	0.9	2.3	2.4	0.9	2.3	2.4
11	0.9	2.4	2.6	0.9	2.3	2.5
12	1.1	2.9	3.1	1.1	2.8	3.0
13	1.2	3.2	3.4	1.0	2.7	2.9
14	1.3	3.5	3.7	1.0	2.5	2.7
15	1.4	3.8	4.0	1.0	2.6	2.8
16	1.2	3.0	3.3	1.1	2.9	3.1
17	1.3	3.3	3.5	1.1	2.8	3.0
18	1.2	3.2	3.4	1.0	2.7	2.9
19	1.2	3.1	3.3	1.2	3.0	3.3
20	1.1	2.9	3.1	1.1	2.8	3.0
21	1.1	2.8	3.0	1.1	2.8	3.0
22	1.3	3.4	3.6	1.0	2.7	2.9
23	1.3	3.3	3.5	1.1	2.8	3.0
24	1.0	2.6	2.8	0.8	2.1	2.3
25	0.9	2.3	2.5	0.8	2.1	2.3
Min	0.8	2.1	2.3	0.7	1.9	2.0
Max	1.4	3.8	4.0	1.3	3.4	3.6
Mean	1.1	2.8	3.0	1.0	2.5	2.7

Table 4. Summary statistics of fluoride hazard quotient.

S. No.	Pre-monsoon			Post-monsoon		
	Adults	Children	Infants	Adults	Children	Infants
1	0.9	2.4	2.5	0.8	2.0	2.1
2	1.1	3.0	3.2	1.0	2.6	2.8
3	1.0	2.5	2.7	0.9	2.4	2.6
4	0.9	2.3	2.4	0.7	1.9	2.0
5	2.1	5.5	5.9	1.3	3.4	3.6
6	1.8	4.8	5.1	0.8	2.1	2.3
7	1.9	5.0	5.3	0.8	2.1	2.3
8	1.4	3.8	4.0	0.8	2.0	2.1
9	1.0	2.5	2.7	0.8	2.1	2.3
10	1.2	3.3	3.5	0.9	2.3	2.4
11	1.1	3.0	3.2	0.9	2.3	2.5
12	1.4	3.8	4.0	1.1	2.8	3.0
13	1.3	3.5	3.7	1.0	2.7	2.9
14	1.2	3.3	3.5	1.0	2.5	2.7
15	1.0	2.5	2.7	1.0	2.6	2.8
16	0.9	2.3	2.4	1.1	2.9	3.1
17	1.0	2.5	2.7	1.1	2.8	3.0
18	1.2	3.3	3.5	1.0	2.7	2.9
19	1.1	3.0	3.2	1.2	3.0	3.3
20	1.5	4.0	4.3	1.1	2.8	3.0
21	1.4	3.8	4.0	1.1	2.8	3.0
22	1.1	3.0	3.2	1.0	2.7	2.9
23	0.7	1.8	1.9	1.1	2.8	3.0
24	0.7	1.8	1.9	0.8	2.1	2.3
25	0.6	1.6	1.7	0.8	2.1	2.3
Min	0.6	1.6	1.7	0.7	1.9	2.0
Max	2.1	5.5	5.9	1.3	3.4	3.6
Mean	1.2	3.1	3.3	1.0	2.5	2.7

samples was executed in the PHREEQC simulation. The results reveal that the precipitation of carbonates, sulphates, and fluorides, as well as the dissolution of chloride minerals, regulate the mechanisms of groundwater chemistry (Fig. 6a- b).

Piper diagram

The results exhibited the dominance of Na-Cl water type, followed by a mixed Ca-Mg-Cl type in both pre- and post-monsoon seasons (Fig. 7). The prevalence of Na-Cl water indicates deeper and more saline groundwater, and the mixed (Ca-Mg-Cl) water could be from the collective impact of anthropogenic and geological controls.

Gibbs and Scatter Plots

The majority of groundwater samples were classified under the rock dominance category. (Fig. 8), which is further supported through the scatter plots (Fig. 9a, c). Geologically, the study area is primarily comprised of shales, with some recent deposits found along the Sagileru, a tributary of the Penna River. These deposits mainly consist of calcite, magnetite, feldspars, clay, and fluorite. Due to silicate weathering and the dissolution of calcite, elements such as Na^+ , K^+ , Ca^{2+} , Mg^{2+} , and F^- may be released into the groundwater (Fig. 9a-c). The Cl^- ions in groundwater may enter the groundwater from rock-water interactions (Fig. 9a). However, NO_3^- may be released into groundwater through natural processes, such as rainfall, oxidation, and reduction, in low quantities. Nevertheless, higher levels of nitrates in water, possibly from anthropogenic inputs like inappropriate waste disposal and intensive agriculture, are further supported by scatter plots (Fig. 9d).

Statistical Analysis

Cluster Analysis

The cluster analysis of ward linkage, using the squared Euclidean distance of observations (sampling locations) in the pre-monsoon

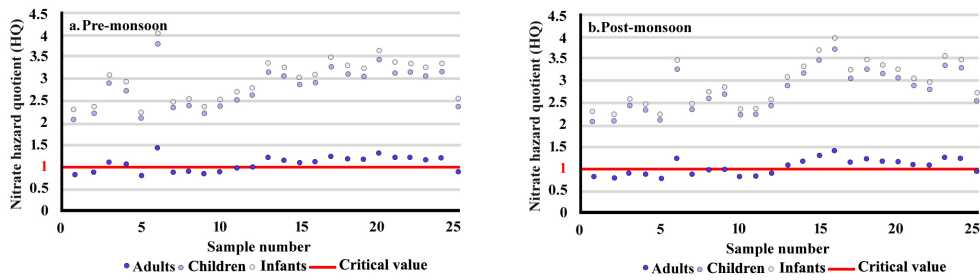


Fig. 4a -b. Nitrate hazard quotient of various age groups.

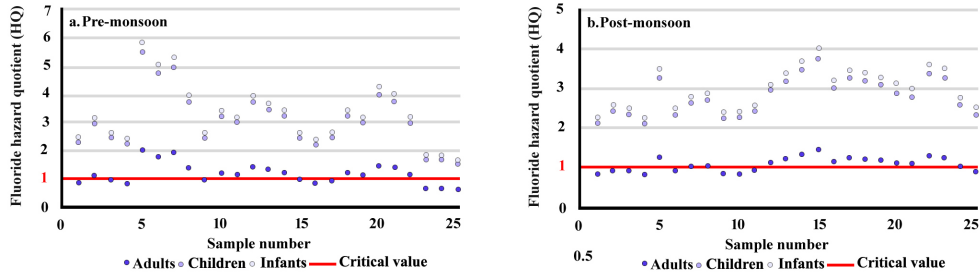


Fig. 5a -b. Fluoride hazard quotient of various age groups.

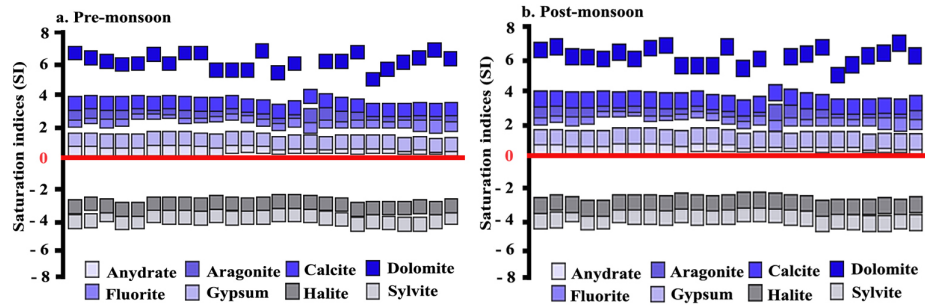


Fig. 6a -b. Temporal trend of saturation indices (SI).

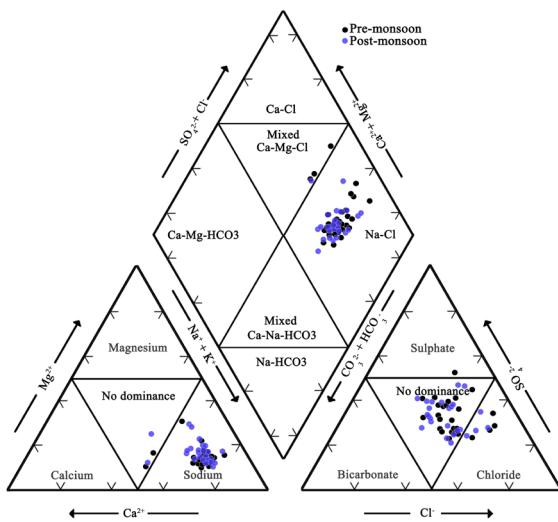


Fig. 7. Piper trilinear diagram showing water facies.

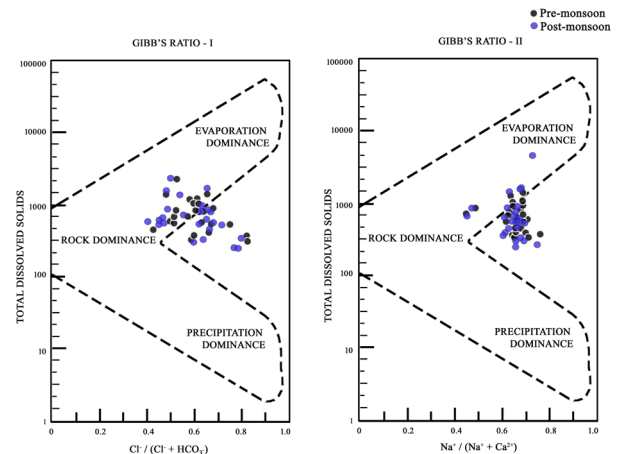


Fig. 8. Gibbs diagram showing controlling mechanisms of groundwater chemistry.

period, yielded three clusters. The first cluster consisted of observations (1, 2, 3, 4, 9, 10 and 11) characterized by low mineralisation with low EC, with low Na^+ , F^- and NO_3^- in concentrations in the groundwater, while cluster 2 comprised observations (5, 6, 7, 8, 17, 18, 19, 20, 21, 22, 23, 24 and 25) that were moderately mineralised with moderate EC, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} and NO_3^- high F^- concentrations

in groundwater. The cluster 3 consisted of observations (12, 13, 14, 15 and 16) that were highly mineralised water with elevated EC, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} , and NO_3^- , as well as low F^- levels in the water (Fig. 10a); whereas in the post-monsoon period, Cluster 1 consisted of observations (1, 2, 3, 5, 6, 7, 8, 18, 19, 20, 21, 22, 23 and 24) which were moderately mineralised with moderate EC, TDS, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} and NO_3^- and high F^- in groundwater. The cluster 2 consisted of observations (4, 9, 10, 11, 15, 16, 17 and

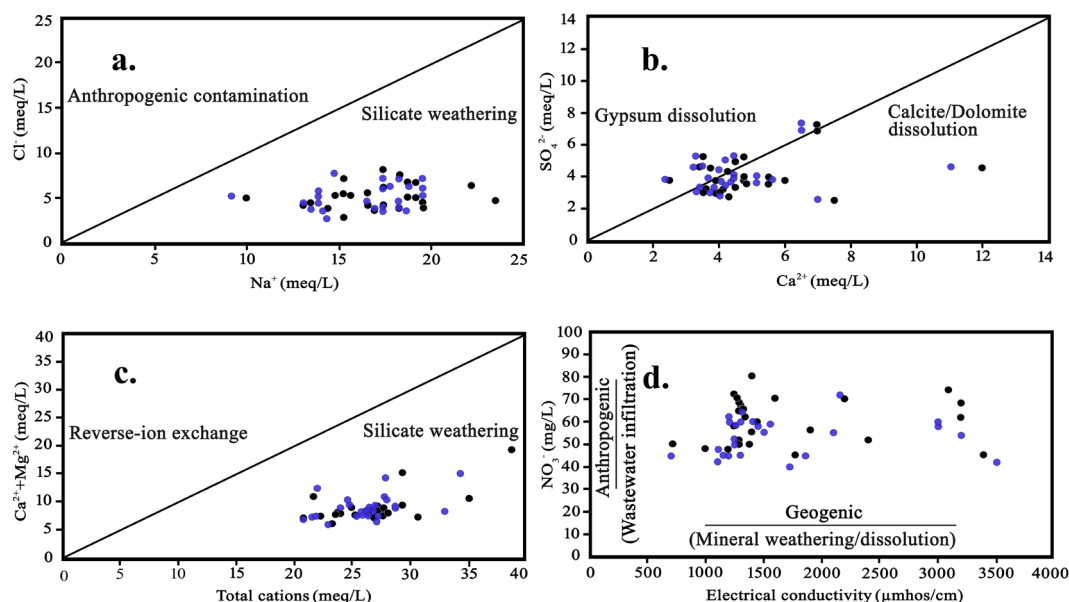


Fig. 9a-d. Scatter plots of (a) Na^+ vs Cl^- , (b) Ca^{2+} vs SO_4^{2-} , (c) Total cations vs $\text{Ca}^{2+} + \text{Mg}^{2+}$ and (d) Electrical conductivity vs NO_3^- .

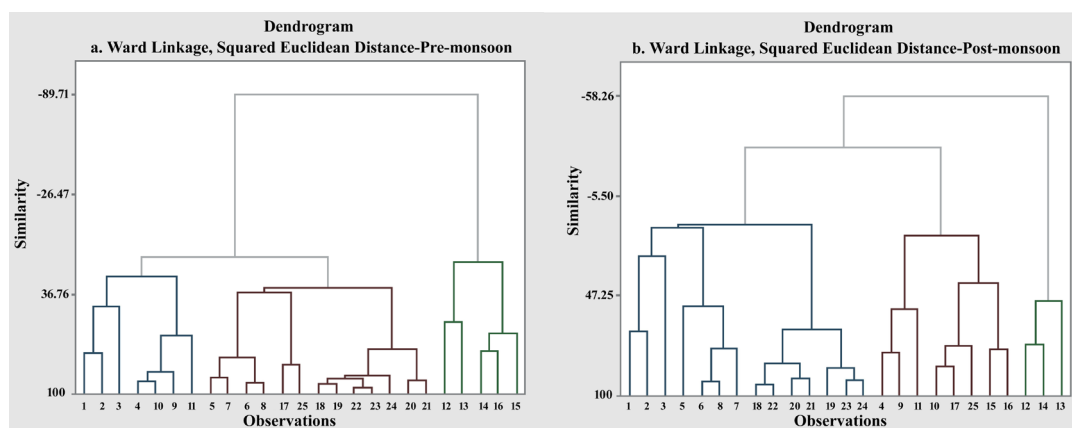


Fig. 10 a-b. Dendrogram showing temporal groundwater clustering patterns.

25) which were low mineralised with low EC, Na^+ , F^- and NO_3^- in concentrations in groundwater, while the cluster 3 consisted of observations (12, 13 and 14) which were highly mineralised with elevated EC, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , SO_4^{2-} and NO_3^- and low to moderate F^- in water (Fig. 10b). However, a few samples shifted from high to moderate mineralisation in the post-monsoon period may be due to the dilution effect of freshwater recharge.

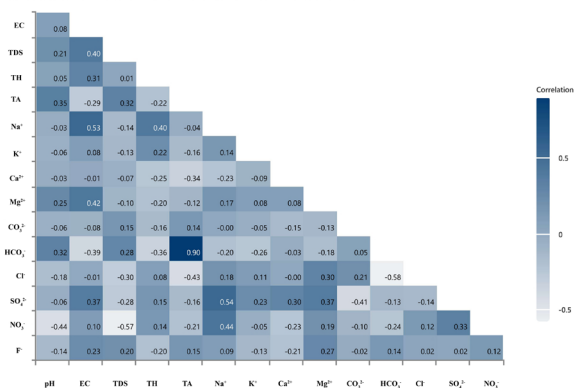
Pearson Correlation Matrix

The pH showed a weak positive correlation with TA and bicarbonates in the pre-monsoon season, demonstrating its weak control in geochemical processes related to TA and bicarbonates (Fig. 11a). EC showed a weak positive correlation with TDS, TH, Mg^{2+} , SO_4^{2-} , and strong moderate correlations with Na^+ , indicating that Na^+ significantly contributes to the overall conductivity than TDS, TH, Mg^{2+} , and SO_4^{2-} (Shrestha and Basnet, 2018). TDS exhibited a weak correlation with TA, indicating a non-linear relationship between TDS and TA (Fig. 11a); TH showed a moderate positive correlation for Na^+ , indicating partial control over TH. (Fig. 11a). TA showed a strong correlation with HCO_3^- indicating that HCO_3^- is the primary contributor to alkalinity in water (Perolo et al., 2023). Na^+ exhibited a moderate positive correlation with SO_4^{2-} , and NO_3^- , suggesting a common origin. The Ca^{2+} showed weak positive correlations with SO_4^{2-} , diverse geochemical processes. Mg^{2+} had a weak positive

correlation with Cl^- , SO_4^{2-} indicating diverse geochemical processes. Moreover, CO_3^{2-} had moderate negative loadings with SO_4^{2-} , indicating the inverse relationship. HCO_3^- exhibited a moderate negative loading with Cl^- , suggesting the contribution of diverse sources. Moreover, SO_4^{2-} exhibited a weak positive correlation with NO_3^- indicating diverse geochemical behaviour (Fig. 11a).

Similarly, the EC exhibited a weak positive correlation with TDS, TH, and SO_4^{2-} , as well as moderate loadings of Na^+ , indicating the partial influence of the aforementioned parameters on EC in the post-monsoon season. (Fig. 11b). TDS exhibited a weak positive correlation with TA, indicating a non-linear relationship between TDS and TA. TH exhibited weak positive correlations with Na^+ , indicating the partial control of Na^+ over TH. TA showed moderate correlations with HCO_3^- , indicating that HCO_3^- is the primary contributor to TA in water. A moderate positive correlation of Na^+ with SO_4^{2-} , suggested a common origin. Mg^{2+} had a weak positive correlation with Cl^- , F^- , indicating diverse geochemical processes. Similar to that in the pre-monsoon period, CO_3^{2-} exhibited moderate negative loadings, with SO_4^{2-} indicating an inverse relationship and HCO_3^- showed a moderate negative loading, with Cl^- indicating the origin of diverse sources. SO_4^{2-} had a weak positive correlation with NO_3^- indicating the diverse geochemical behaviour. Both monsoons showed nearly similar results. However, the shifting of moderate to weak correlations for EC, TDS, TA, Mg^{2+} , K^+ , SO_4^{2-} and NO_3^- in the

a. Pearsons correlation matrix-Pre-monsoon



b. Pearsons correlation matrix-Post-monsoon

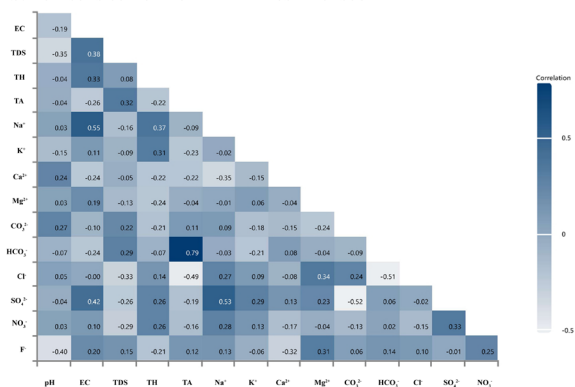


Fig. 11 a-b. Temporal variation of analysed parameters using the Pearson correlation matrix.

post-monsoon season could be attributed to the dilution effect of freshwater recharge.

CONCLUSIONS

The current study highlights the significance of various physicochemical parameters, including TDS, Ca²⁺, Mg²⁺, Na⁺, K⁺, Cl⁻, NO₃⁻, and F⁻, in groundwater during both the pre- and post-monsoon seasons. It reveals that the groundwater in Badvel, India, is influenced by both natural and human-induced factors. Specifically, Ca²⁺, Mg²⁺, Na⁺, K⁺, Cl⁻, and F⁻ can be attributed to silicate weathering, while SO₄²⁻ and NO₃⁻ are attributed to human-induced sources, such as agricultural runoff, leachates from municipal effluents, and untreated sewage. Consequently, Na-Cl emerges as the dominant water facies, highlighting the interconnectedness between these ionic contributions and the resultant water chemistry. Additionally, the precipitation of carbonate, sulphates, and fluoride, along with the dissolution of chloride minerals, plays a crucial role as major controlling factors in groundwater dynamics. Based on PIG, the majority of the groundwater samples (90%) were categorised as low pollution. However, in contrast, a significant proportion of the samples revealed concerning levels of nitrate and fluoride pollution, with "infants followed by children and adults exhibiting values exceeding >1 HQ during both monsoon seasons, making the water unsafe for human consumption. Notably, the post-monsoon period water showed lower PIG and HQ values, likely due to the dilution effects from rainfall recharge compared to pre-monsoon period water.

The current study underscores the need for further research to investigate the long-term trends of groundwater quality in Badvel, India, particularly the impact of agricultural practices and urban development on water chemistry. Future investigations should focus

on providing proper waste-disposal facilities and on the effectiveness of proposed treatment methods, such as bio-adsorbents for defluoridation and advanced filtration techniques for nitrate removal to mitigate contamination and enhance water safety for communities reliant on this resource.

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Data Availability: To ensure privacy, security and minimise bias, we provide it upon request only.

References

- Adimalla, N. (2019) Spatial distribution, exposure, and potential health risk assessment from nitrate in drinking water from semi-arid region of South India. *Hum. Ecol. Risk Assess.*, v.26(2), pp.310–334. <https://doi.org/10.1080/10807039.2018.1508329>.
- APHA (American Public Health Association). (2023) *Standard Methods for the Examination of Water and Wastewater*. American Public Health Association: Washington, DC, USA, ISBN 978-0-87553-299-8.
- Benavente, D., Ruiz, M., García-Martínez, N., Vergara, M., Fernandez-Cortes, A. and Sanchez-Moral, S. (2025) Estimation of mineral saturation and CO₂ partial pressure in natural waters using electrical conductivity and pH: A fast and versatile on-site tool. *J. Hydrol.*, v.660, pp.1–14. <https://doi.org/10.1016/j.jhydrol.2025.133474>.
- BIS (Bureau of Indian Standards). (2003) *Drinking water - specification*, Bureau of Indian Standards: New Delhi IS; 10500.
- Brindha, K., Rajesh, R., Murugan, R. and Elango, L. (2012) Nitrate pollution in groundwater in some rural areas of Nalgonda district, Andhra Pradesh, India. *J Environ Sci Eng.*, v.54(1), pp.64–70.
- CGWB (Central Ground Water Board). (2025) *Aquifer Mapping for Sustainable Management of Ground Water Resources In Y.S.R Kadapa District, Andhra Pradesh*. Central Groundwater Board, Southern Region, Hyderabad, pp.1–52.
- Das, S. (2022) Groundwater management in India: some recent breakthroughs. *J. Geol. Soc. India.*, v.98(2), pp.151–154. <https://doi.org/10.1007/s12594-022-1950-0>.
- Das, S. (2023) Groundwater sustainability, security and equity: India today and tomorrow. *J. Geol. Soc. India.*, v.99(1), pp.5–8. <https://doi.org/10.1007/s12594-023-2260-x>.
- Gugulothu, S., Rao, N.S., Das, R.K. and Dhakate, R. (2022) Geochemical evaluation of groundwater and suitability of groundwater quality for irrigation purposes in an agricultural region of South India. *Appl Water Sci.*, v.12(6), pp.1–13. <https://doi.org/10.1007/s13201-022-01583-w>.
- Gibbs, R.J. (1970) Mechanisms controlling world water chemistry. *Sci.*, v.170(3962), pp.1088–1090. <https://doi.org/10.1126/science.170.3962.1088>.
- Kharaka, Y.K., Abedini, A.A., Gans, K.D., Thordson, J.J., Beers, S.R. and Thomas, R.B. (2018) Changes in the chemistry of groundwater reacted with CO₂: Comparison of results from laboratory experiments and the ZERT field site, Bozeman, Montana, USA. *Appl. Geochem.*, v.98, pp.75–81. <https://doi.org/10.1016/j.apgeochem.2018.08.017>.
- Li, P. and Wu, J. (2019) Drinking water quality and public health. *Expo. Health.*, v.11, pp.73–79. <https://doi.org/10.1007/s12403-019-00299-8>.
- Madhav, S., Raju, N.J., Ahamad, A., Singh, A.K., Ram, P. and Gobel, W. (2021) Hydrogeochemical assessment of groundwater quality and associated potential human health risk in Bhadohi environs, India. *Environ. Earth Sci.*, v.80(17), pp.1–14. <https://doi.org/10.1007/s12665-021-09824-y>.
- Perolo, P., Escoffier, N., Chmiel, H.E., Many, G., Bouffard, D. and Perga, M. (2023) Alkalinity contributes at least a third of annual gross primary

- production in a deep stratified hardwater lake. *Limnol. Oceanogr.*, v.8(2), pp.359–367. <https://doi.org/10.1002/lol2.10311>.
- Piper, A.M. (1944) A graphic procedure in the geochemical interpretation of water-analyses. *Eos, Transactions American Geophysical Union*, v.25(6), pp.914-928. <https://doi.org/10.1029/TR025i006p00914>.
- Rajmohan, N. and Elango, L. (2006) Hydrogeochemistry and its relation to groundwater level fluctuation in the Palar and Cheyyar river basins, southern India. *Hydrol. Process.*, v.20(11), pp.2415–2427. <https://doi.org/10.1002/hyp.6052>.
- Reddy, C.C., K.V.C., Golla, V., Badapalli, P.K. and Reddy, N.B.Y. (2022) Evaluation of groundwater contamination for fluoride and nitrate in Nellore Urban Province, Southern India: a special emphasis on human health risk assessment (HHRA). *Appl Water Sci.*, v.12(32), pp.1–9. <https://doi.org/10.1007/s13201-021-01537-8>.
- Reddy, M.R., Raju, N.J., Reddy, Y.V. and Reddy, T.V.K. (2000) Water resources development and management in the Cuddapah district, India. *Environ. Geol.*, v.39(3–4), pp.342–352. <https://doi.org/10.1007/s002540050013>.
- Santaiah, M., Sudhakar, G. and Kamakshamma, J. (2020) Determination of groundwater quality in the Badvel Region, Kadapa, Andhra Pradesh, India. *Int. J. Res. Dev. Res. Technol.*, v.8(2), pp.124–128.
- Saravanan, K., Srinivasamoorthy, K., Gopinath, S., Prakash, R. and Suma, C.S. (2016) Investigation of hydrogeochemical processes and groundwater quality in Upper Vellar sub-basin Tamil Nadu, India. *Arab. J. Geosci.*, v.9(5), pp.1–14. <https://doi.org/10.1007/s12517-016-2369-y>.
- Shaji, E., Gómez-Alday, J., Hussein, S., Deepu, T.R. and Anilkumar, Y. (2018) Salinization and deterioration of groundwater quality by nitrate and fluoride in the Chittur Block, Palakkad, Kerala. *J. Geol. Soc. India.*, v.92(3), 337–345. <https://doi.org/10.1007/s12594-018-1017-4>.
- Shaji, E., Sarath, K., Santosh, M., Krishnaprasad, P., Arya, B. and Babu, M.S. (2023) Fluoride contamination in groundwater: A global review of the status, processes, challenges, and remedial measures. *Geosci Front.*, v.15(2), pp. 1–29. <https://doi.org/10.1016/j.gsf.2023.101734>.
- Shrestha, A.K. and Basnet, N. (2018) The correlation and regression analysis of physicochemical parameters of river water for the evaluation of percentage contribution to electrical conductivity. *J. Chem.*, v.2018(1), pp.1–9. <https://doi.org/10.1155/2018/8369613>.
- Sirigala, L., Ramani, P., G.P. and Ramalingam, K. (2023) Prevalence of dental and skeletal fluorosis among school children in rural areas of YSR Kadapa District, Andhra Pradesh, India. *Cureus.*, v.15(12), pp.1–13. <https://doi.org/10.7759/cureus.51288>.
- Sreedevi, P.D., Sreekanth, P.D., Shakeel, Ahmed. and Reddy, D.V. (2019) Evaluation of groundwater quality for irrigation in a semi-arid region of South India. *Sustain. Water Resour. Manag.*, v.5(3), pp.1043–1056. <https://doi.org/10.1007/s40899-018-0279-8>.
- Subba Rao, N. (2012) PIG: a numerical index for dissemination of groundwater contamination zones. *Hydrol. Process.*, v.26(22), pp.3344–3350. <https://doi.org/10.1002/hyp.8456>.
- Sunitha, V. and Muralidhara Reddy, B. (2018) Defluoridation of water using *Mentha longifolia* (Mint) as Bioadsorbent. *J. Ind. Geophys. Union.*, v.22(2), pp.207–211.
- Tiwari, V. and Rao, N.S. (2025) Impact of climate change on water resources in India. *J. Geol. Soc. India.*, v.101(6), pp.974–977. <https://doi.org/10.17491/jgsi/2025/174194>.
- USEPA (United States Environmental Protection Agency). (2015) USA National Primary Drinking Water Regulations. <http://water.epa.gov/drink/contaminants/upload/mcl-2.pdf>. (accessed 7/9/2015).
- Yarlanki, V.K., Vadithya, G.N., Etikala, B. and Golla, V. (2025) Hydrogeochemical characterization and source evaluation of groundwater contamination in parts of the Uddanam CKDU area, Southern India: integrating statistical, geochemical modeling, pollution, and health indices. *Int. J. Environ. Res.*, v.19(3), pp.1–24. <https://doi.org/10.1007/s41742-024-00730-2>.



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